The impact of e-banking service quality on e-customer satisfaction: Evidence from the undergraduates of the University of Ruhuna.

A research paper submitted to the Department of Mathematics of University of Ruhuna in partial fulfillment of the requirements for the award of the Honours Degree of Bachelor of Science in Financial Mathematics and Industrial Statistics

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Abstract

As a customer expectation continue to rise, the quality of e-banking services has become a pivotal factor in determining customer satisfaction. This study investigates the impact of e-banking service quality on e-customer satisfaction, as existing literature present inconclusive findings. This highlights the need for an empirical investigation to bridge the research gap. This study explores the relationship between service quality and e-customer satisfaction, aiming to identify the key dimensions of service quality that influence customer perceptions and loyalty in the digital banking landscape.

Using a structured questionnaire, data was collected from a 104 sample of e-banking users who were randomly selected from the FMIS second year undergraduate students. and quantitative analysis was conducted to examine the relationship between service quality dimensions such as reliability, responsiveness, web design, functionality, privacy and security and overall customer satisfaction.

The findings reveal that service quality significantly impacts e-customer satisfaction, with security and ease of use emerging as the most influential factors. The study provides valuable insights for banking institutions to enhance their e-banking platforms, improve customer experiences .Finally the outcomes reveal that there has a positive relationship between service quality and e-customer satisfaction.

Introduction

E-banking service quality is a key factor in customer satisfaction and the success of electronic banking services. Some dimensions of e-banking service quality such as reliability, responsiveness, security and privacy, web design and functionality are mainly affected to the customer satisfaction. Customer satisfaction is a measure of how happy customers are with a company's products, services, and capabilities. It's important for business success because satisfied customers are more likely to remain loyal, make repeat purchases, and refer others to the business.

High-quality e-banking services are essential for ensuring customer satisfaction. As technology advances, maintaining high service quality has become a essential thing for financial institutions in a competitive market. In the other hand, E-customer satisfaction is the degree to which customers are content with their overall experience when using e-banking services. Satisfied e-customers are mostly loyal to the bank, and recommend the platform to others.As more customers turn to online banking for convenience and efficiency, the quality of these services

becomes the primary factor in determining whether customers will continue to use the platform or switch to competitors. Because of that identifying this relationship is very useful.

Also, the relationship between e-banking service quality and e-customer satisfaction is significant for both academic, research, and practical applications. Especially the nature of this relationship remains unclear in certain demographic groups, particularly among university students. This study conducts to fill this gap by investigating the relationship between e-banking service quality and e-customer satisfaction among undergraduates at the University of Ruhuna. For this purpose, we will conduct a survey and analyze main dimensions of service quality. And we believe, the findings of this study will offer valuable insights for banks in tailoring their e-banking services to better meet the needs of undergraduate students.

RESEARCH QUESTIONS

1. What is the impact of e-banking service quality on e-customer satisfaction?
2. What is the impact of reliability on e-customer satisfaction?
3. What is the impact of responsiveness on e-customer satisfaction?
4. What is the impact web design on e-customer satisfaction?
5. What is the impact of functionality on e-customer satisfaction?
6. What is the impact of privacy and security on e-customer satisfaction?

OBJECTIVE OF STUDY

1. To identify the impact of e-banking service quality on e-customer satisfaction.
2. To identify the impact of reliability on e-customer satisfaction.
3. To identify the impact of responsiveness on e-customer satisfaction.
4. To identify the impact web design on e-customer satisfaction.
5. To identify the impact of privacy and security on e-customer satisfaction.
6. To identify the impact of functionality on e-customer satisfaction.

Literature review

This literature review aims to synthesize the existing research findings on the impact of e banking service quality on e-customer satisfaction.

E banking service have become essential for customers in managing their financial transactions conveniently and efficiently. As banking sector become more competitive, understanding the impact of service quality on customer satisfaction has become crucial. E- customer satisfaction refers to the overall evaluation of an e-banking service based on various factors such as reliability, functionality, responsiveness, security, privacy, and web design. A well-developed understanding of these factors can significantly enhance the user experience and foster loyalty in a competitive banking environment. This section reviews the literature related to the five dimensions of e-banking service quality and their impact on customer satisfaction. or customers to adopt the e-banking practices. The study conducted by Nuradhi K. Jayasiri, W.A.S.P.Weerathunga (2008) investigated The Popularity of E banking in Sri Lanka. They found that majority of the customers in Sri Lanka were aware of e banking facilities, but most of them had not tried using them. but most of them had not tried using them. They still preferred traditional banking methods such as paying bills, withdrawing money, and depositing cheques at bank counters. State banks had lower usage of e-banking facilities compared to other banks.

Reliability in e-banking refers to the consistency and dependability of the banking services, ensuring that transactions are processed accurately and timely. Reliability is crucial for building trust and satisfaction in e-banking systems (Sathiyavany & Shivany, 2018). Research by Keramati and Nili (2011) highlights that a reliable system is fundamental for customer satisfaction, especially when users are engaged in time-sensitive tasks such as fee payments or fund transfers. In their study on customer relationship management (CRM) and retention, they found that e-banking services that operate without interruption and with high accuracy significantly contribute to customer loyalty. Beigi et al. (2016) also argue that reliable service quality in e-banking positively impacts customer satisfaction, particularly in the context of users seeking seamless experiences in online transactions. When university students or young users rely on digital banking, their expectation for consistent, error-free service is high.

Functionality in e-banking refers to the breadth and depth of services offered by the platform, such as bill payments, money transfers, loan applications, and account balance management. Functionality is crucial for customer satisfaction, as it determines whether a customer can perform all desired tasks efficiently (Sathyabama & Samundeswari, 2019). Ayinaddis et al. (2023) argue that a well-functioning platform that supports a variety of essential services enhances

customer satisfaction, especially when the platform's functionality aligns with the users' financial needs. For university students, whose activities are often time-sensitive, the availability of necessary features such as quick fund transfers or instant bill payments can directly influence their satisfaction. Othman (2014) also notes that mobile banking functionality is a major factor driving adoption among Generation Y, reinforcing the importance of an extensive range of services.

Responsiveness is defined as the ability of a bank to effectively and efficiently address customer inquiries, issues, and service requests. Fast response times and quick issue resolution significantly contribute to customer satisfaction in e-banking services (Sathiyavany & Shivany, 2018). Khan et al. (2022) found that responsive customer service, particularly through various channels such as live chat, email, or phone, enhances users' experiences with e-banking platforms. Keramati and Nili (2011) also observed that customers who receive timely assistance, especially when facing challenges like transaction errors, report higher satisfaction levels. In the case of university students, where time constraints are common, prompt customer service is even more critical. Students expect a fast response, particularly when they need immediate support for critical transactions like fee payments.

Web design plays a significant role in user satisfaction, as it directly influences the ease of use and navigation of e-banking platforms. A well-designed interface ensures that customers can efficiently complete their banking tasks without confusion or frustration (Beigi et al., 2016). Sathiyavany and Shivany (2018) argue that a user-friendly web design that includes clear navigation tools, appealing visuals, and straightforward functionalities enhances the overall user experience, ultimately leading to greater customer satisfaction. Ayinaddis et al. (2023) found that users are more likely to trust e-banking platforms that feature clean, intuitive, and aesthetically pleasing designs. This is particularly important for younger customers, such as university students, who expect modern, visually attractive, and easy-to-navigate digital interfaces. Lelasari and Bernarto (2023) further assert that an intuitive web design leads to improved customer loyalty, as users find the e-banking platform more convenient and enjoyable to use.

The dimensions of **security and privacy** are perhaps the most crucial when considering customer satisfaction in e-banking services. As students and other users manage sensitive financial data online, they demand a high level of security for their personal and financial information. According to Lee et al. (2011), perceived security plays a significant role in shaping customer satisfaction. Users are less likely to feel satisfied with e-banking services if they feel that their data is at risk of fraud or unauthorized access. Chellappa and Sin (2005) emphasize that privacy concerns, particularly in how personal data is collected and shared, can influence customer trust. Hammoud et al. (2018) assert that when e-banking platforms have transparent

security measures and adhere to strict privacy policies, they foster trust among their customers. Khan et al. (2022) also highlight that banks focusing on the privacy of customer data, while implementing encryption and secure login protocols, improve user confidence and satisfaction.

In the context of university students, who are typically more digitally savvy and conscious of data protection, secure banking environments and transparent privacy policies are essential for improving satisfaction levels. The assurance of secure transactions coupled with clear privacy protections builds a sense of safety and trust, leading to greater customer satisfaction.

Customers satisfaction and service quality

Service quality and satisfaction are constructs derived from comparing expectations and performance, but they are not necessarily equivalent. Perceived service quality is an attitude and long-term evaluation, while customer satisfaction is transaction-specific. The study conducted by Jyotsna Sharma, Jagdeep Singh, Amandeep Singh (2020) explored the Impact of E-Banking Service Quality on Customer Satisfaction. The study found that e-banking service quality has a significant impact on customer satisfaction. Nine dimensions of service quality were identified, including tangibility, credibility, communication, security, and responsiveness, which were found to have a significant effect on customer satisfaction. The regression model used in the study was a reasonable fit, indicating a statistically significant association.

between service quality dimensions and customer satisfaction. The findings highlight the importance of offering quality services in the banking industry, particularly in a developing country like India, where customers have diverse needs and expectations. The study emphasizes the role of customer satisfaction in driving repeat usage of e-banking services. A.P.P. Perera, H.M.S. Priyanath (2018) further analyzed the impact of internet banking service quality on customer satisfaction using Partial Least Squares-Structural Equation Modeling (PLS-SEM). The reliability and validity of six dimensions of customer satisfaction were evaluated, and all indicators met the condition of reliability and had statistically significant factor loadings. The study found that efficiency, system availability, and compensation had a positive and significant impact on customer satisfaction, while fulfillment, privacy, and responsiveness did not have a significant impact. Contact was also found to have a positive influence on customer satisfaction. The study concluded that internet banking service quality dimensions have a higher influence on customer satisfaction. The measurement model and structural model were assessed using PLS-SEM, and collinearity issues were checked.

RESEARCH METHODOLOGY

RESEARCH PARADIGM

This study adopts a **positivist research paradigm**, which is ideal for examining the impact of e-banking service quality on e-customer satisfaction. Positivism assumes that reality is objective and measurable, aligning with the study's goal of quantifying relationships between variables like reliability, responsiveness, and satisfaction.

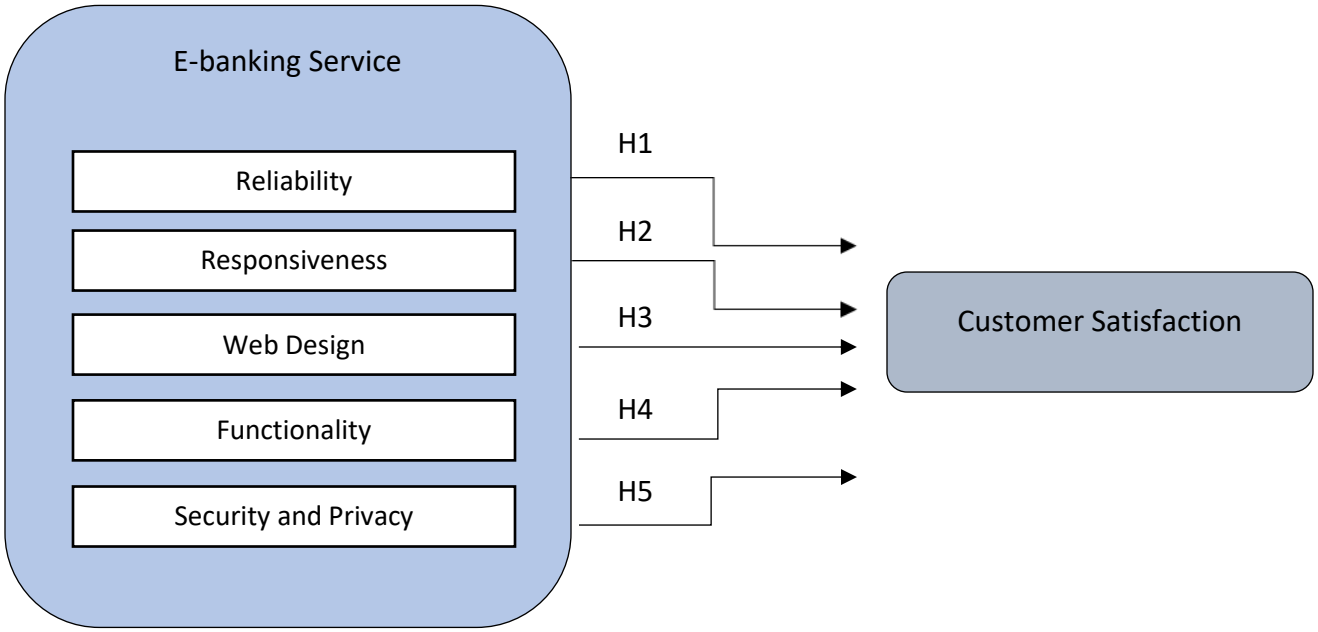
Ontological Assumption that, Positivism views reality as singular and independent of the researcher. In this study, e-banking service quality and e-customer satisfaction are treated as observable, measurable phenomena that can be analyzed objectively. Epistemological Assumption that, Knowledge is gained through objective observation and measurement. Data will be collected via structured surveys to measure undergraduates' perceptions, focusing on identifying causal relationships and generalizable patterns. Methodological Approach that, A quantitative research design is used, consistent with positivism. Statistical tools like regression and correlation analysis will test hypotheses, ensuring a systematic and empirical examination of the research problem.

The researcher remains objective and detached, collecting and analyzing data without bias. This ensures findings are based solely on measurable evidence, maintaining the study's integrity.

Positivism aims to generalize findings to broader populations. By studying undergraduates at the University of Ruhuna, insights may apply to other universities or young e-banking users.

Positivism is the most suitable paradigm for this study because it focuses on measuring and quantifying the impact of specific e-banking service quality dimensions on customer satisfaction. The use of a structured, quantitative approach ensures that the findings are reliable, replicable, and generalizable to a wider population. Additionally, the emphasis on hypothesis testing and statistical analysis is consistent with the positivist goal of identifying universal laws or patterns.

CONCEPTUAL FRAMEWORK



HYPOTHESIS

H1: Reliability positively influences e-customer satisfaction.
Reliability refers to the consistency and dependability of an e-service in delivering promised outcomes. When customers perceive an e-platform as reliable, they are more likely to trust it, leading to higher satisfaction. For instance, accurate order fulfillment, timely delivery, and error-free transactions enhance customer trust and satisfaction. This hypothesis aligns with the expectation that reliable services reduce uncertainty and foster positive customer experiences.

H2: Responsiveness positively influences e-customer satisfaction.
Responsiveness measures how quickly and effectively an e-service addresses customer inquiries, complaints, or issues. Faster response times and efficient problem resolution are critical in the digital environment, where customers expect immediate attention. High responsiveness demonstrates a company's commitment to customer care, which can significantly enhance satisfaction levels and build long-term loyalty.

H3: Web design positively influences e-customer satisfaction.
Web design encompasses the visual appeal, usability, and navigation of an e-commerce platform. A well-designed website that is intuitive, aesthetically pleasing, and easy to navigate enhances the user experience. When customers find it easy to locate products, access information, and complete transactions, their overall satisfaction increases. This hypothesis highlights the importance of user-centric design in driving positive customer perceptions.

H4: Privacy and security positively influences e-customer satisfaction.
Privacy and security are critical concerns for e-customers, especially with the increasing prevalence of data breaches and cyber threats. When customers feel that their personal and financial information is secure, they are more likely to trust the platform and engage in repeat transactions. Strong privacy policies, secure payment gateways, and transparent data handling practices contribute to higher satisfaction by alleviating customer concerns about potential risks.

H5: Functionality positively influences e-customer satisfaction.
Functionality refers to the technical performance and features of an e-commerce platform, such as site speed, mobile compatibility, and seamless integration of services. A platform that functions smoothly without glitches or downtime provides a hassle-free experience, which is essential for customer satisfaction. This hypothesis emphasizes the role of technical excellence in meeting customer expectations and ensuring a positive online shopping experience.

RESEARCH METHODOLOGY

The study adopts a quantitative research design to explore the relationship between e-banking service quality and customer satisfaction focused on the undergraduate students of Financial Mathematics and Industrial Statistics degree program in University of Ruhuna who actively use e-banking services. Primary data method is used to collect data by using the questionnaire survey over a sample of 104 students from the FMIS second year students to ensure reliability and statistical validity. Secondary data will be drawn from the relevant research papers on the impact of e-banking service quality on customer satisfaction. The research will be conducted at University of Ruhuna. We plan to use online via Google Form / questioner to collect data and it will be analyzed using SPSS software. The data is analyzed using descriptive statistics, correlation analysis, and regression analysis using **SPSS** software to explore relationships and impacts and the findings are presented using graphs, tables, and charts to illustrate trends and relationships clearly.

Descriptive Statistics

	N	Mean	Std. Deviation
I can find what I need on the e-banking site/app without any difficulty	104	2.2788	.79387
Do you trust e-banking services for the accurate and efficient completion of your financial transactions.	104	1.8942	.62190
The e-banking system performs as expected without errors	104	2.3942	.87487
The response time of the e-banking platform is fast and efficient.	104	2.1923	.76421
I am able to easily reach support when needed through e-banking channels	104	2.1250	.78441
If I encounter an issue, the customer support team responds promptly.	104	2.3942	.89679
The design of the e-banking website is visually appealing and easy to navigate.	104	2.1923	.75140
The information on the e-banking website is organized in a clear and understandable way.	104	2.1538	.76029
I can find what I need on the e-banking site/app without any difficulty.	104	2.1250	.78441

The e-banking services offer all the features I need (e.g., transfers, bill payments, etc.).	104	2.1442	.71622
I am able to perform transactions without encountering any technical issues.	104	2.2692	.82710
The e-banking platform works smoothly across different devices (e.g., mobile, laptop, etc.).	104	2.0577	.81045
I feel that my personal data is protected when using e-banking services.	104	2.0865	.72555
I am confident that e-banking services use secure methods for transactions	104	2.0000	.77585
The e-banking platform clearly communicates its privacy policies and data protection practices.	104	2.0962	.75709
Overall, I am satisfied with the quality of e-banking services.	104	2.1058	.76219
I would recommend using e-banking services to other university students.	104	2.0769	.69243
I believe the e-banking services I use meet my expectations in terms of convenience and reliability.	104	2.1154	.77969
Valid N (listwise)	104		

Reliability Analysis

The reliability analysis conducted for our research article reveals varying levels of internal consistency across the different scales used in the study. For the reliability, the Cronbach's Alpha value of 0.549 indicates moderate reliability, suggesting that the scale has a reasonable level of internal consistency. Similarly, the analysis for the responsiveness yields a Cronbach's Alpha value of 0.543, also reflecting moderate reliability. Both scales demonstrate acceptable consistency, though there may be potential for further refinement to enhance their reliability. All 104 cases for these analyses were valid, with no exclusions, confirming the completeness of the data

VARIABLE	NO. OF MEASUREMENT ITEMS	CRONBACH'S ALPHA VALUE
Reliability	3	0.549
Responsiveness	3	0.543
Web Design	3	0.404
Functionality	3	0.431
Security and Privacy	3	0.510
E-Customer Satisfaction	3	0.579

In contrast, the analysis for the Web Design scale, results in a Cronbach's Alpha value of 0.404, indicating low reliability. This suggests that the scale may not be consistently measuring the intended construct, and the items may require revision or replacement to improve internal consistency. Similarly, the analysis for the functionality shows a Cronbach's Alpha value of 0.431, reflecting low to moderate reliability. While this scale performs slightly better than the previous one, it still indicates room for improvement to achieve higher consistency. Both analyses were conducted on 104 valid cases, with no exclusions, ensuring the integrity of the data.

Finally, the analysis for the Security and Privacy yields a Cronbach's Alpha value of 0.510, indicating moderate reliability. This suggests that the scale has a reasonable level of internal consistency, though further refinement could potentially enhance its reliability. Overall, the findings highlight the varying degrees of reliability across the scales used in the study, with some demonstrating moderate reliability and others showing lower reliability. These results underscore the importance of continued scale development and validation to ensure robust measurement of the constructs under investigation. All analyses were conducted on complete datasets with 104 valid cases and no exclusions, ensuring the reliability of the findings.

Analysis for Validity.

VARIABLE	KMO VALUE	Approx. Chi-Square	DF	SIG:
Reliability	0.620	25.494	3	0.000
Responsiveness	0.604	23.776	3	0.000
Web Design	0.560	10.697	3	0.013
Functionality	0.530	15.343	3	0.002
Security and Privacy	0.587	20.462	3	0.000
E-Customer Satisfaction	0.631	27.398	3	0.000

The factor analysis conducted for our research article reveals varying levels of construct validity across the different scales. For most dimensions, the Kaiser-Meyer-Olkin (KMO) values range from 0.530 to 0.631, indicating that the data is generally suitable for factor analysis. Bartlett's Test of Sphericity is significant for all dimensions ($p < 0.05$), confirming the appropriateness of the analysis. The communalities range from 0.223 to 0.639, and the extracted components explain between 45.800% and 54.375% of the total variance, suggesting moderate to strong construct validity for most dimensions. Factor loadings range from 0.472 to 0.799, indicating that the items generally contribute well to the underlying constructs. However, some dimensions, particularly those with lower communalities and variance explained, may require further refinement to improve their validity. Overall, the results support the use of these dimensions in the study, though continued development is recommended for enhanced measurement robustness.

Testing Hypothesis

HYPOTHESIS	DIMENSION	PEARSON CORRELATION	SIG:(2 TAILED)
H1	Reliability	0.384	0.000
H2	Responsiveness	0.223	0.023
H3	Functionality	0.389	0.000
H4	Security and Privacy	0.363	0.000
H5	Satisfaction	0.438	0.000

The correlation analysis examines the relationships between five dimensions (Reliability, Responsiveness, Web Design, Functionality, and Security) and Satisfaction. Here's a summary of the correlations between each dimension and Satisfaction:

1. Testing the Hypothesis (H1)

There is a moderate positive correlation between Reliability and Satisfaction, which is statistically significant at the 0.01 level. This suggests that higher perceived reliability of e-banking services is associated with higher satisfaction levels.

2. Testing the Hypothesis (H2

There is a weak positive correlation between Responsiveness and Satisfaction, which is statistically significant at the 0.05 level. This indicates that better responsiveness in e-banking services is linked to higher satisfaction, though the relationship is not as strong as with Reliability.

3. Testing the Hypothesis (H3)

There is a moderate positive correlation between Web Design and Satisfaction, which is statistically significant at the 0.01 level. This suggests that a well-designed and user-friendly e-banking website is associated with higher satisfaction levels.

4. Testing the Hypothesis (H4)

There is a moderate positive correlation between Functionality and Satisfaction, which is statistically significant at the 0.01 level. This indicates that the functionality of e-banking services (e.g., ease of use, features) is positively related to user satisfaction.

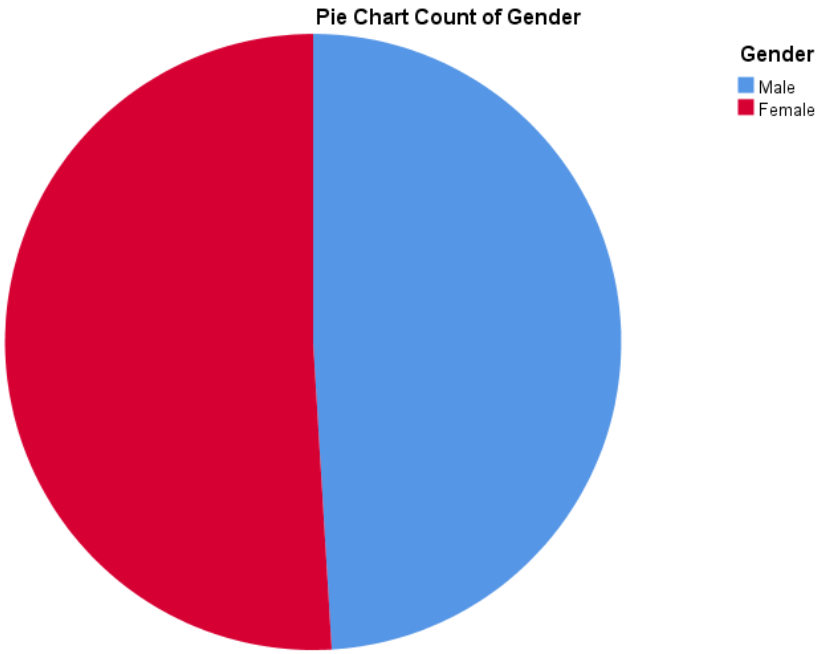
5. Testing the Hypothesis (H5)

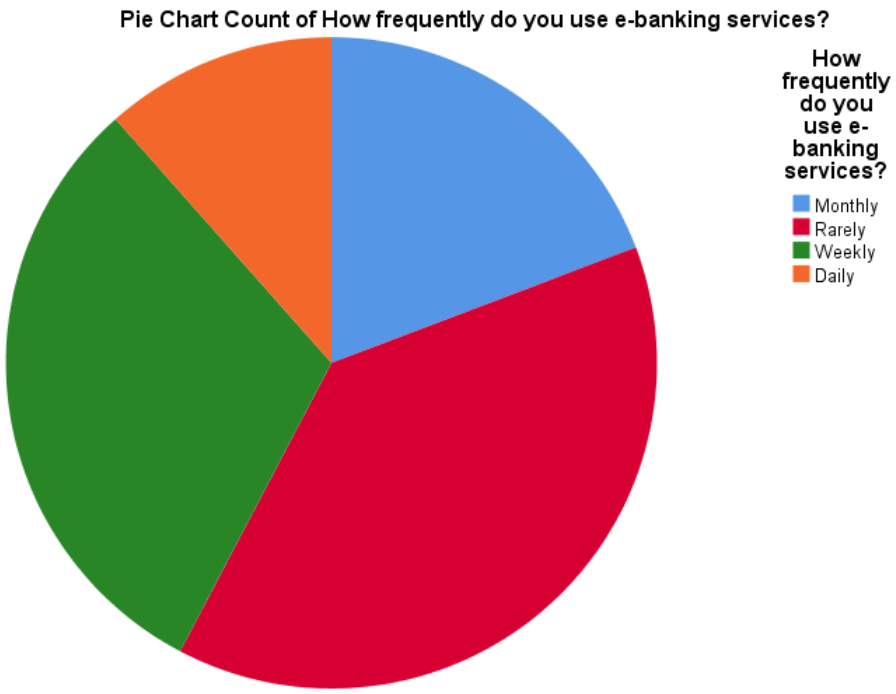
There is a moderate to strong positive correlation between Security and Satisfaction, which is statistically significant at the 0.01 level. This suggests that higher perceived security in e-banking services is strongly associated with higher satisfaction levels.

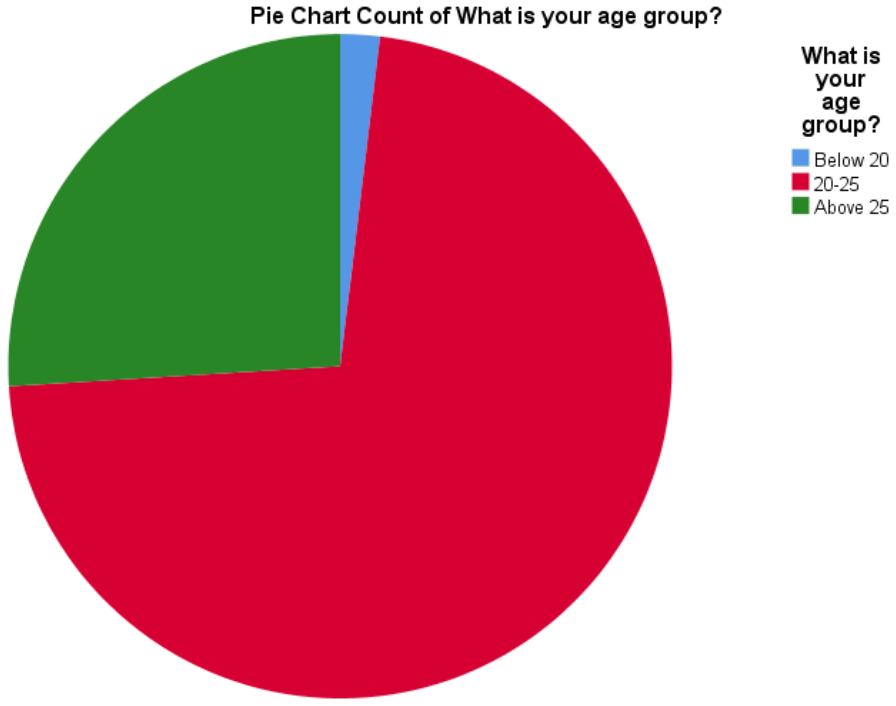
All five dimensions (Reliability, Responsiveness, Web Design, Functionality, and Security) are positively correlated with Satisfaction, with Security showing the strongest relationship. This indicates that improving these dimensions, particularly Security, could lead to higher user satisfaction with e-banking services.

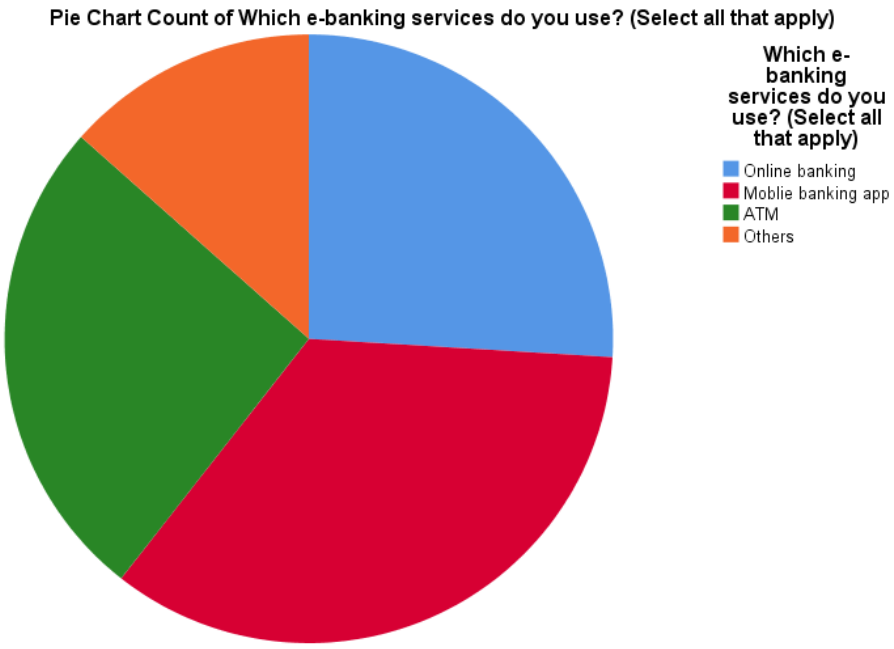
FREQUENCIES OF DEMOGRAPHICAL INFORMATION

E-banking has become an integral part of modern financial transactions, and this analysis explores its usage among a sample of 104 respondents based on SPSS frequency distribution results. The gender distribution in the sample is nearly equal, with 51% female and 49% male respondents. The majority of participants (72.1%) belong to the 20-25 age group, while 26% are above 25 years old, and only 1.9% are below 20, indicating that young adults are the primary users of e-banking services in this study. In terms of usage frequency, 38.5% of respondents reported using e-banking services rarely, making it the most common response, followed by 30.8% who use it weekly, 19.2% monthly, and only 11.5% daily, showing that frequent usage remains relatively low. Among the different services available, mobile banking apps are the most preferred, with 34.6% of respondents using them, while online banking and ATM services are equally popular at 26% each, and 13.5% reported using other e-banking services. The preference for mobile banking apps reflects a shift towards digital transactions via smartphones, emphasizing the importance of user-friendly and secure digital banking platforms. These findings suggest that while e-banking is accessible, factors such as convenience, security, and awareness may influence its adoption and frequent use. To enhance engagement, financial institutions should focus on improving accessibility, security, and user experience to encourage more consistent utilization of e-banking services.




Pie Chart Count of How frequently do you use e-banking services?
How frequently do you use e-banking services?
Monthly
Rarely
Weekly
Daily


Pie Chart Count of What is your age group?
What is your age group?
Below 20
20-25
Above 25



Scale Selection(questionnaire)

Variable		Questions
Reliability	RELI1	1. The e-banking service is consistently available without unexpected downtimes.
	RELI2	2. Do you trust e-banking services for the accurate and efficient completion of your financial transactions.
	RELI3	3. The e-banking system performs as expected without errors.
Responsiveness	RESP1	4. The response time of the e-banking platform is fast and efficient.
	RESP2	5. I am able to easily reach support when needed through e-banking channels.
	RESP3	6. If I encounter an issue, the customer support team responds promptly.

Web Design	WEBD1	7. The design of the e-banking website is visually appealing and easy to navigate.
	WEBD2	8. The information on the e-banking website is organized in clear and understandable way.
	WEBD3	9. I can find what I need on the e-banking site/app without any difficulty.
Functionality	FUN1	10. The e-banking services offer all the features I need (e.g., transfers, bill payments, etc.).
	FUN2	11. I am able to perform transactions without encountering any technical issues.
	FUN3	12. The e-banking platform works smoothly across different devices (e.g., mobile, laptop, etc.).
Security and Privacy	SECU1	13. I feel that my personal data is protected when using e-banking services.
	SECU2	14. I am confident that e-banking services use secure methods for transactions.
	SECU3	15. The e-banking platform clearly communicates its privacy policies and data protection practices.
E-Customer Satisfaction	SATIS1	16. Overall, I am satisfied with the quality of e-banking services.
	SATIS2	17. I would recommend using e-banking services to other university students.
	SATIS3	18. I believe the e-banking services I use meet my expectations in terms of convenience and reliability.

CONCLUSION

This study aimed to investigate the impact of e-banking service quality on e-customer satisfaction among FMIS undergraduate students at the University of Ruhuna. The findings revealed that service quality dimensions such as reliability, responsiveness, web design, functionality, and security and privacy significantly influence e-customer satisfaction. Among these, security and privacy emerged as the most critical factors, followed by functionality and reliability. This indicates that students place a high value on the safety of their personal and financial information, as well as the ease of use and reliability of e-banking platforms.

The study also highlighted that while e-banking services are widely accessible, their frequent usage remains relatively low among students. Mobile banking apps were the most preferred mode of e-banking, reflecting a shift towards digital transactions via smartphones. This underscores the importance of developing user-friendly, secure, and reliable e-banking platforms to enhance customer satisfaction and encourage more consistent usage.

The findings provide valuable insights for banking institutions to improve their e-banking services, particularly in terms of security, functionality, and user experience. By addressing these key areas, banks can better meet the needs of their customers, particularly the younger, tech-savvy demographic represented by university students.

SUGGESTIONS

Future research should aim to expand the sample size and diversity by including participants from different age groups, educational backgrounds, and geographic locations. This would help generalize findings to a broader population and provide more robust insights.

Incorporating mixed-methods approaches, such as combining surveys with interviews or focus groups, could offer deeper insights into customer perceptions and experiences with e-banking services. This approach would enrich the understanding of how users interact with digital banking platforms.

Additionally, future studies should explore other dimensions of service quality, such as customer support, transaction speed, and advanced features. These factors could provide a more comprehensive understanding of what drives customer satisfaction in e-banking.

Conducting cross-cultural studies would also be valuable, as it would help understand how cultural differences influence perceptions of e-banking service quality and satisfaction. This is particularly relevant for global banks operating in diverse markets.

Finally, adopting a longitudinal approach could help track changes in customer satisfaction over time, providing insights into how improvements in service quality impact customer loyalty and retention in the long run.

OUTCOMES

The study highlights that security and privacy are the most critical factors for e-customer satisfaction, urging banks to implement robust measures like encryption and transparent policies. Functionality and reliability also significantly impact satisfaction, emphasizing the need for user-friendly platforms with essential features and minimal technical issues. Mobile banking apps are preferred, indicating the importance of seamless mobile experiences.
While responsiveness and web design have a moderate impact, improving customer support and intuitive navigation remains vital. Despite accessibility, frequent e-banking usage is low among students, suggesting a need for increased awareness and promotion. These insights help banks enhance digital services, boost satisfaction, and gain a competitive edge.

REFERENCES

Sathiyavany N, Shivany S (2018), E-Banking Service Qualities, E-Customer Satisfaction, and e-Loyalty: A conceptual Model, The International Journal of Social Sciences and Humanities Invention, vol. 5, Issue 06, Jun, 2018

Abbas Keramati, Alireza Nili (2011), A Proposal Framework for investigating the impact of customer relationship management on customer retention in ecommerce, Proceedings of the 2011 International Conference on Industrial Engineering and Operations Management, Kuala Lumpur, Malaysia, January 22 – 24, 2011

Ali Naghi Mansour Beigi1, Saeid Jorfi, Reza Salemi Tajarrod, Hamed Mansour Beigi (2016), The Study of the Effects of E-banking Services Quality on Customers' Satisfaction and Loyalty (Case Study: Agricultural Bank of Khuzestan Province), journal of current research in science issn 2322-5009 coden (usa):

Adilah Binti Othman (2014), factors influencing the adoption of mobile banking among generation y
Samuel Godadaw Ayinaddis, Birhan Ambachew Taye,Bantie Gentnet Yirsaw (2023), Examining the effect of electronic banking service quality on customer satisfaction and loyalty: an implication for technological innovation.

nuradhi k. jayasiri, w.a.s.p. weerathunga (2008), popularity of e-banking in sri lanka
Jyotsna Sharma, Jagdeep Singh, Amandeep Singh (2020), Impact of E-Banking Service Quality on Customer Satisfaction, International Journal of Recent Technology and Engineering (IJRTE) ISSN: 2277 3878, Volume-8 Issue-5, January 2020.

Dr.K.Sathyabama, R.Samundeswari (2019), a study on customer satisfaction towards internet banking services in thoothukudi city.

Nur Lelasari, Innocentius Bernarto (2023-02-01), The Impact of Customer Satisfaction on Customer Loyalty of BCA Mobile Banking.

Shahid Kalim Khan, Najam-Ul-Hassan, Muhammad Naeem Anjum (2022), factors influencing customer satisfaction in e-banking services in Pakistan: evidence based on reliability, privacy & convenience
Liss Jenneboer , Carolina Herrando (2022), The Impact of Chatbots on Customer Loyalty: A Systematic Literature Review.

Jamil Hammoud, Rima M. Bizri, and Ibrahim El Baba (2018), The Impact of E-Banking Service Quality on Customer Satisfaction: Evidence from the Lebanese Banking Sector.

Do Thanh, Van Thanh, Dung Manh, Duyen Bich T (2020), Impact of Service Quality, Customer Satisfaction and Switching Costs on Customer Loyalty, Journal of Asian Finance, Economics and Business Vol 7 No 8 (2020) 395–405.

A.P.P. Perera, H.M.S. Priyanath (2018), The impact of internet banking service quality on customer Satisfaction, International Journal of Management, IT & Engineering, Vol. 8 Issue 2, February 2018.